

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (MS CS, 2.5+ years) deploying LLM-powered workflows in regulated enterprise environments with full compliance accountability. Expert in prompt engineering, RAG systems, and AI-enabled workflow configuration for high-stakes, audit-sensitive contexts. Delivered 35% retrieval precision improvement and HIPAA-grade data governance infrastructure for healthcare AI - directly applicable to CLM/legal AI regulatory review and contract analysis. Proven client-facing communicator across technical and non-technical enterprise stakeholders. Remote-ready.

CORE COMPETENCIES

LLM Workflow Configuration • Prompt Engineering & Tuning • RAG (Retrieval-Augmented Generation) • AI Output Evaluation & Validation • Compliance-Grade Data Governance • REST API Integration • Python / FastAPI / Docker • Enterprise Client Deployment • Audit Trails & Explainability • HIPAA / Regulated Industry AI

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting, SaaS Healthcare

- Configured and deployed **AI-enabled LLM workflows** for healthcare enterprise; tuned prompts and evaluated model outputs for reliability; improved stability ~**25%**, reduced hallucinations >**30%**.
- Built **RAG systems** for clinical knowledge retrieval; retrieved precision improved ~**35%**; retrieval-grounded response alignment exceeded **90%** in evaluation.
- Validated and refined AI-driven outputs for **HIPAA compliance**: de-identification, data lineage documentation, evaluation audit trails, stakeholder-facing system-limitation docs.
- Translated real-world use cases (eligibility checks, care workflow navigation, clinical documentation) into implemented LLM solutions using FastAPI/Flask RESTful services.
- Deployed via **Docker containers**; integrated with enterprise REST APIs; reduced post-deploy defects ~**30%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Applied NLP

- RAG document Q&A for non-technical staff: reduced hallucinations ~**30%**, raised accuracy >**85%**. Flask API deployment.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- Enterprise SaaS SQL optimization for Order-to-Cash platform. RBAC and audit logging for financial modules in compliance-sensitive oil & gas environment. Jenkins CI/CD automation.

KEY PROJECTS

Agentic Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline with schema-validated agent contracts, **RAG-grounded regulatory validation** (CPT/ICD policy documents), and explainable risk scoring with **audit-ready reasoning traces**. Built for compliance-sensitive healthcare domain - direct parallel to Agiloft's contract analysis AI with regulatory review and auditability requirements.

Dream Decoder - AI Workflow Configuration & Prompt Tuning

FastAPI + React app orchestrating Perplexity Sonar + Replicate. Configured intermediate prompt transformation layers; ~**30% contextual alignment improvement**. Demonstrates applied prompt engineering for production AI workflow configuration.

Patient Records Management Web Application

Bottle + SQLite healthcare CRUD app with normalized schema, foreign-key constraints, and aggregate queries. Application-layer data governance for patient/diagnosis records. Enterprise SaaS implementation analog.

EDUCATION

Master of Science, Computer Science - Kent State University, Ohio

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, SQL, JavaScript

LLM & AI: RAG, prompt engineering, LangChain, LlamaIndex, vector search, agentic workflows, structured outputs

APIs & Integration: REST, JSON, FastAPI, Flask, Docker

Domains: Healthcare AI (HIPAA-conscious), regulated enterprise SaaS, compliance-grade audit trails

Work Authorization: F-1 OPT (US-authorized now; future sponsorship required)